Genuine Multipartite Entanglement Analysis

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Model
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Hubbard ring (1-D, PBC)

System parameters
-----------------
Sites (L)         : 4
Electrons (Ne)    : 4
Hopping (t)       : 1.0
Interaction (U)   : 2.0
Mode              : dimerized
n-body order      : 3
```

```
Ground-state properties
-----------------------
E₀                : -2.47213595
E₁                : -1.23606798

⟨S²⟩              : 0.000000
S                 : 0.0000
⟨Sz⟩              : 0.000000
Multiplicity 2S+1 : 1.0
Spin state        : Singlet (S=0)

Double occupancy
----------------
⟨D⟩ total         : 0.552786
⟨D⟩ / site        : 0.138197
```

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GME summary
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Extracted n-spin RDM : n = 3

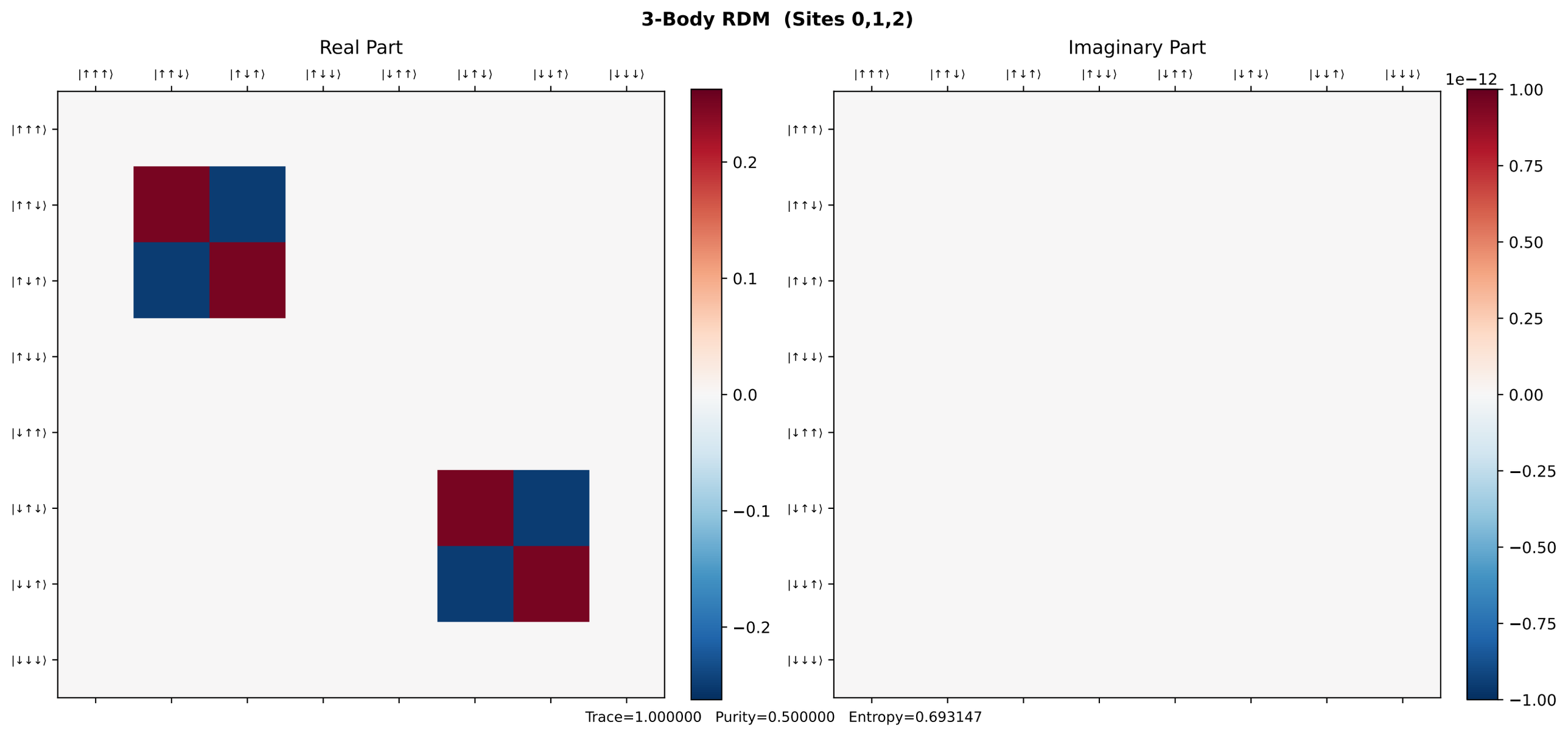
Purity:
  γ = Tr(ρ²)

Separable cut criterion:
  γ_A = 1  ↔  ρ = ρ_A ⊗ ρ_B

C_GME (Troiani et al.):
  C_GME = √(2 · min_A (1 − Tr(ρ_A²)))

Decision logic:
  γ = 1  and  all Tr(ρ_A²) < 1   → GME
  γ = 1  and  some Tr(ρ_A²) = 1  → No GME
  γ < 1                           → Mixed (convex-roof needed)

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3-Body RDM (Sites 0,1,2)
Real Part
Imaginary Part
|↑↑↑⟩ |↑↑↓⟩ |↑↓↑⟩ |↑↓↓⟩ |↓↑↑⟩ |↓↑↓⟩ |↓↓↑⟩ |↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=1.000000 Purity=0.500000 Entropy=0.693147

```
RDM Metrics  (Sites = 0,1,2)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.5000000000
  Von Neumann entropy    : 0.6931471806
  Eigenvalue  min        : 0.000000e+00
  Eigenvalue  max        : 5.000000e-01
```

GME Analysis — 3-Body RDM (Sites 0,1,2)

Bipartitions (Sites 0,1,2)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.50000000$

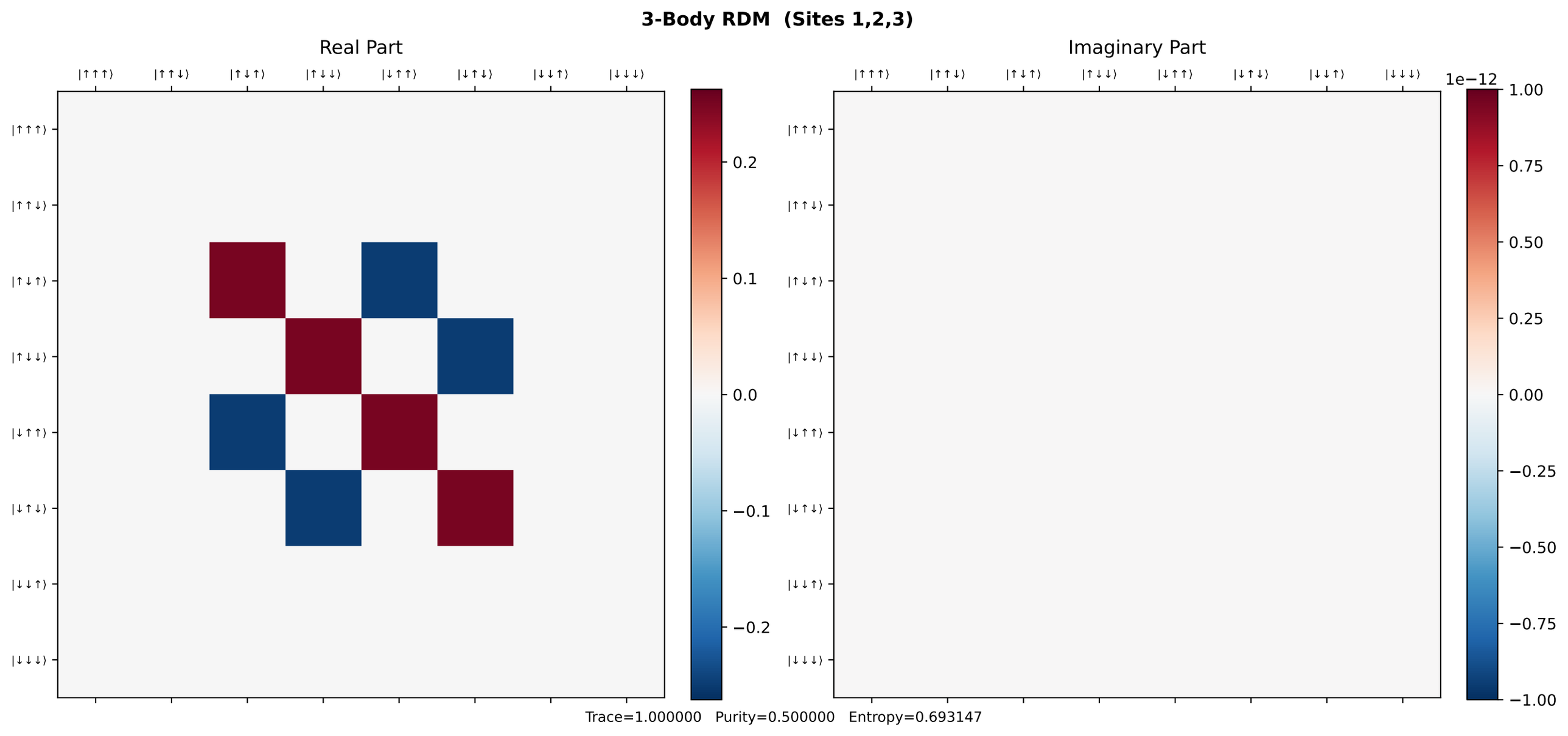
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


3-Body RDM (Sites 1,2,3)
Real Part
Imaginary Part
|↑↑↑⟩ |↑↑↓⟩ |↑↓↑⟩ |↑↓↓⟩ |↓↑↑⟩ |↓↑↓⟩ |↓↓↑⟩ |↓↓↓⟩
0.2 0.1 0.0 −0.1 −0.2
1e−12
1.00 0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75 −1.00
Trace=1.000000 Purity=0.500000 Entropy=0.693147

```
RDM Metrics  (Sites = 1,2,3)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.5000000000
  Von Neumann entropy    : 0.6931471806
  Eigenvalue  min        : -2.834637e-17
  Eigenvalue  max        : 5.000000e-01
```

GME Analysis — 3-Body RDM (Sites 1,2,3)

Bipartitions (Sites 1,2,3)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.50000000$

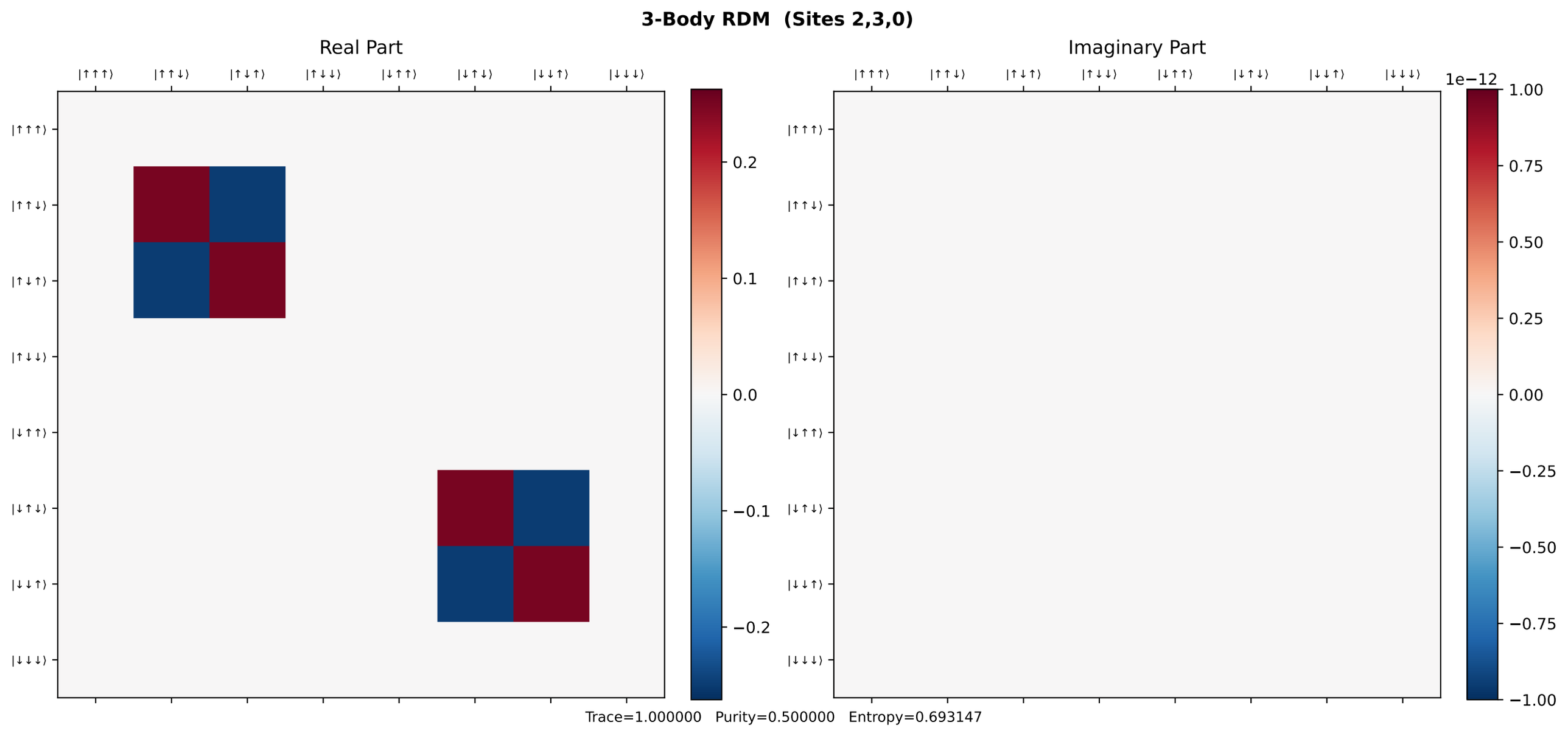
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


3-Body RDM (Sites 2,3,0)
Real Part
Imaginary Part
|↑↑↑⟩ |↑↑↓⟩ |↑↓↑⟩ |↑↓↓⟩ |↓↑↑⟩ |↓↑↓⟩ |↓↓↑⟩ |↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=1.000000 Purity=0.500000 Entropy=0.693147

```
RDM Metrics  (Sites = 2,3,0)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.5000000000
  Von Neumann entropy    : 0.6931471806
  Eigenvalue  min        : -2.775558e-17
  Eigenvalue  max        : 5.000000e-01
```

GME Analysis — 3-Body RDM (Sites 2,3,0)

Bipartitions (Sites 2,3,0)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.50000000$

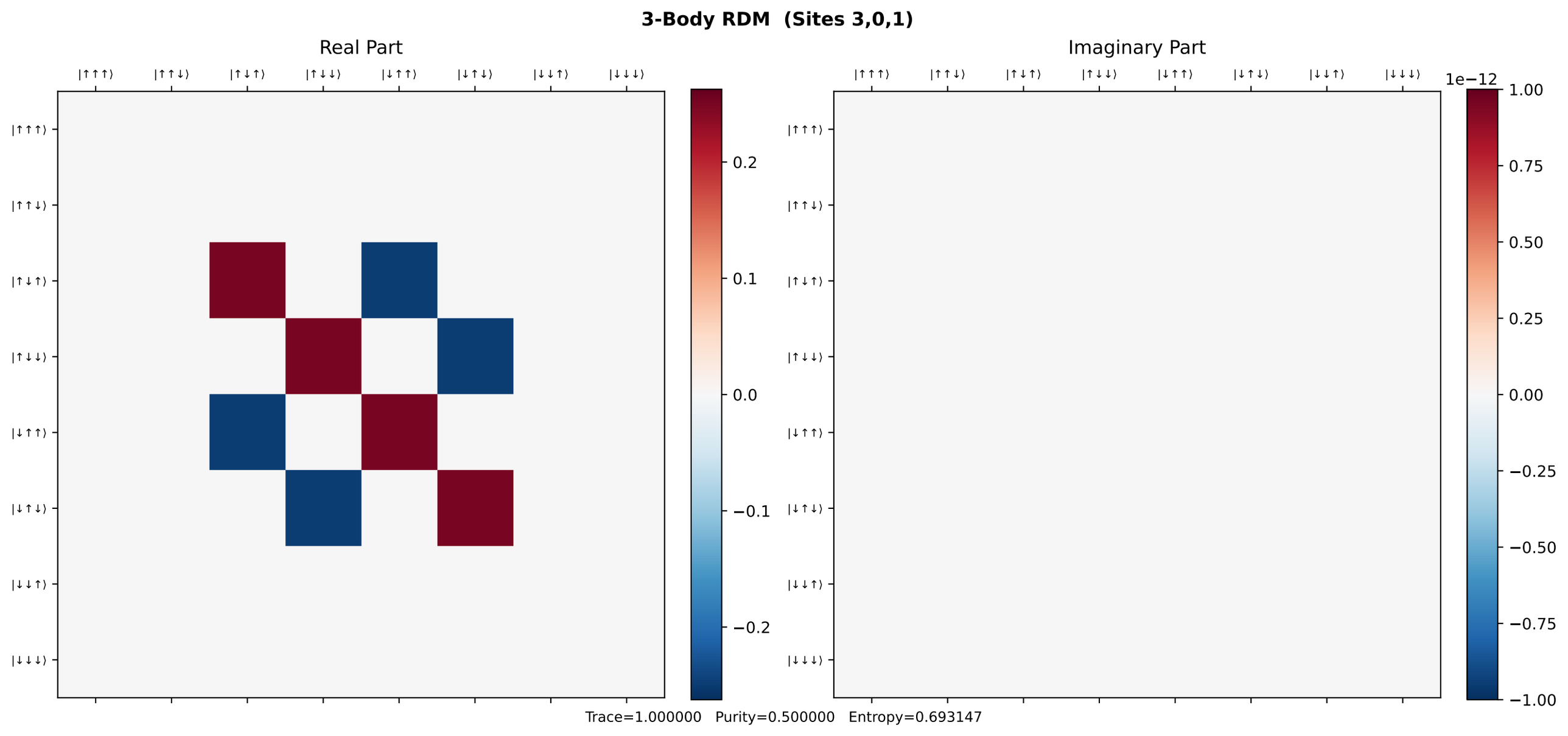
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


3-Body RDM (Sites 3,0,1)
Real Part
Imaginary Part
|↑↑↑⟩ |↑↑↓⟩ |↑↓↑⟩ |↑↓↓⟩ |↓↑↑⟩ |↓↑↓⟩ |↓↓↑⟩ |↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=1.000000 Purity=0.500000 Entropy=0.693147

```
RDM Metrics  (Sites = 3,0,1)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.5000000000
  Von Neumann entropy    : 0.6931471806
  Eigenvalue  min        : -2.775558e-17
  Eigenvalue  max        : 5.000000e-01
```

GME Analysis — 3-Body RDM (Sites 3,0,1)

Bipartitions (Sites 3,0,1)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.50000000$

A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).